

Report R20 — the growth sequences have names: twenty-three OEIS identifications

Krylov sector analysis of quantum cellular automata

Tier 1 analysis

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Abstract

Every growth law in this project was derived here — from an exact linear recurrence over the wall grammar (R18), from a closed form (R19), or, in the cases R16 flagged, from a fit. That fixes an exponent but says nothing about whether the underlying integer sequence is *known*. We asked the On-Line Encyclopedia of Integer Sequences: **24 identifications, covering 38 (rule, observable) pairs and drawing on 18 distinct entries**, each matched as an exact contiguous block of 17 terms with the index shift recorded. Three of the identifications change what we can say. (i) The sequence that *falsifies* the task’s φ prediction for rule 28 is [A023434](#), “dying rabbits” — Fibonacci’s rabbits with a finite lifetime — so the falsification is a classical combinatorial object rather than a numerical accident, and rule 70 lands on the same entry by independent measurement. (ii) W_{156}/W_{198} ’s $D_{\max}$ is [A178715](#), the “Select All, Copy, Paste” problem, whose cost structure (a factor m costs $m+1$) is exactly why the base is $4^{1/5}$, and whose own recurrence $a(n) = 4a(n-5)$ reproduces our ladder term for term. (iii) W_{108} ’s frozen-state count is [A006498](#), whose entry states the Fibonacci-product factorisation R19 derived. Two further identifications are new results in their own right: W_{105} ’s $D_{\max}$ is the central binomial coefficient [A001405](#), and W_{150} ’s is [A214282](#), the largest Euler characteristic of a downset on the N -cube. We also record what is *absent*: the four pbc series of the dissipative ρ -group return no hits at all, so they appear to be new.

Contents

1	Why bother	2
2	Method, and what “matched” means	2
2.1	Our side	2
2.2	No sequence is inherited from a symmetry partner	2
2.3	The OEIS side	3
3	The identifications	3
4	Three that are more than bookkeeping	5
4.1	Dying rabbits, and rule 28	5
4.2	Select All, Copy, Paste, and the $4^{N/5}$	5
4.3	W_{108} ’s frozen states, factorised by the entry itself	6
5	Two identifications that are new results	6
6	What is not there	6
7	Two junk hits, so that they are not rediscovered	7

1 Why bother

An OEIS identification is not a theorem, and none of the physics below depends on one. It buys three specific things.

An external oracle. An entry carries a b-file with hundreds of terms. Our frontier stops at $N = 17$ for the dissipative rules and $N = 22$ for the unitary ones; the named sequence does not. When the planned extension runs on larger hardware, the first hundred terms of the continuation are already written down by someone who has never seen this code. That is the only kind of check on a growth law that is not produced by the thing it checks.

A name for a falsification. R9’s task sec. 6 predicted $a_{\text{wcc}} \geq \varphi$ for rule 28, on the grounds that it inherits rule 156’s 01 wall grammar. It does not hold, and `test_wcc.py`’s `test_rule_28_sector_count_follows_the_plastic_number_not_phi` pins why: the `obc0` sector count obeys the exact recurrence

$$a(N) = a(N - 1) + a(N - 2) - a(N - 4),$$

whose characteristic polynomial factors as $(x - 1)(x^3 - x - 1)$, giving the plastic number $\rho = 1.324718$ — well below $\varphi = 1.618034$. That recurrence is the *definition* of [A023434](#). Section 4.1 takes this further than a coincidence.

A mechanism for an exponent. $4^{1/5} = 1.319508$ is an odd-looking constant to fall out of a cellular automaton. [A178715](#) explains it, and Section 4.2 spells the explanation out.

2 Method, and what “matched” means

2.1 Our side

Unitary series come from `scaling.paper_figures`’s merged Tier-1a + Tier-1e loader, $N = 6 \dots 22$ (R19). Dissipative series are assembled by `scaling.oeis.diss_series`: the Tier-1a archive for $N \geq 6$, and a fresh recomputation from `graph.scc` for $N \leq 12$, with the overlap *asserted* to agree rather than assumed. The recomputation is what supplies $N = 1 \dots 5$, and those low terms are exactly what fixes an index shift, so they had to come from a code path whose agreement with the archive is checked.

Two conventions matter. The ergodic early-exit is switched off during the recomputation, so that a small unit reports its class count instead of bailing out — with it on, W_{73} reads $0, 0, 0, 4, \dots$ instead of $1, 2, 3, 4, \dots$ and the shift comes out wrong by three. And D_{max} for a dissipative rule means the largest *recurrent* class, i.e. the largest key of `size_hist`, not the largest weakly connected component; the two differ (for W_{70} at $N = 12$ they are 16 and 486) and only the former is the object R-T14 says the renewal counts.

2.2 No sequence is inherited from a symmetry partner

Several identifications cover more than one rule. In every such case the rules were measured separately and found to agree; none is copied across a symmetry. This is not pedantry. `core.rules` warns that reflection is *not* a guaranteed sector-size symmetry for dissipative rules under the even-first convention, so $W_{28} = W_{70}$ and $W_{29} = W_{71} = W_{157} = W_{199}$ are measurements, not shortcuts. `test_oeis.py::test_grouped_rules_were_each_measured_not_inherited` recomputes each member of each group and compares.

2.3 The OEIS side

Entries are fetched once and cached in `analytics/oeis_terms.json`, so the tests run offline and a failure means our series moved rather than that OEIS did. A match requires that *every* term we have appears *consecutively and in order* inside the entry's listed terms; nothing is matched on a truncated prefix, up to a constant factor, or on a subsequence. The shift is then read off against the entry's declared offset and stored, so that

$$\text{ours}(N) = \text{Axxxxxx}(N + \text{shift})$$

is a checkable statement and not a gesture. All 23 currently verify on 17 terms each.

3 The identifications

rules	bc	series	OEIS	index	terms	N	growth
W_{28}, W_{70}	obc0	n_{rec}	A023434	$a(N+2)$	17	1–17	1.324718
$W_{29}, W_{71}, W_{157}, W_{199}$	obc0	n_{rec}	A000931	$a(N+6)$	17	1–17	1.324718
W_{73}	obc0	n_{rec}	A000930	$a(N+1)$	17	1–17	1.465571
W_{109}	obc0	n_{rec}	A179070	$a(N+2)$	17	1–17	1.465571
$W_{28}, W_{29}, W_{70}, W_{71}, W_{157}, W_{199}$	obc0	D_{max}	A173862	$a(N+2)$	17	1–17	1.259921
W_{73}	obc0	D_{max}	A000045	$a(N+2)$	17	1–17	1.618034
W_{109}	obc0	D_{max}	A000045	$a(N+0)$	17	1–17	1.618034
W_{73}, W_{109}	pbc	n_{rec}	A001609	$a(N+0)$	17	1–17	1.465571
W_{73}, W_{109}	pbc	D_{max}	A169985	$a(N+0)$	17	1–17	1.618034
W_{108}	obc0	n_{rec}	A005251	$a(N+3)$	17	6–22	1.754878
W_{201}	obc0	n_{rec}	A005251	$a(N+2)$	17	6–22	1.754878
W_{156}, W_{198}	obc0	n_{rec}	A000045	$a(N+2)$	17	6–22	1.618034
W_{150}	obc0	n_{rec}	A008619	$a(N+1)$	17	6–22	1.000000
W_{105}	obc0	n_{rec}	A004525	$a(N+2)$	17	6–22	1.000000
W_{108}	obc0	D_{max}	A000045	$a(N+0)$	17	6–22	1.618034
W_{201}	obc0	D_{max}	A000045	$a(N+2)$	17	6–22	1.618034
W_{156}, W_{198}	obc0	D_{max}	A178715	$a(N+0)$	17	6–22	1.319508
W_{150}	obc0	D_{max}	A214282	$a(N+2)$	17	6–22	2.000000
W_{105}	obc0	D_{max}	A001405	$a(N+1)$	17	6–22	2.000000
W_{108}	obc0	n_{froz}	A006498	$a(N+2)$	17	6–22	1.618034
W_{201}	obc0	n_{froz}	A195971	$a(N-1)$	17	6–22	1.618034
W_{156}, W_{198}	obc0	n_{froz}	A008619	$a(N+1)$	17	6–22	1.000000
W_{150}	obc0	n_{froz}	A000034	$a(N-6)$	17	6–22	1.000000
W_{105}	obc0	n_{froz}	A007877	$a(N-5)$	17	6–22	1.000000

and the entries themselves, with the closed form in our own index where we have one:

OEIS	entry name	closed form in our N
A023434	Dying rabbits: $a(n) = a(n-1) + a(n-2) - a(n-4)$.	$a(N) = a(N-1) + a(N-2) - a(N-4)$
A000931	Padovan sequence (or Padovan numbers): $a(n) = a(n-2) + a(n-3)$ with $a(0) = 1, a(1) = a(2) = 0$.	$a(N) = a(N-2) + a(N-3)$
A000930	Narayana's cows sequence: $a(0) = a(1) = a(2) = 1$; thereafter $a(n) = a(n-1) + a(n-3)$.	$a(N) = a(N-1) + a(N-3)$
A179070	$a(1)=a(2)=a(3)=1, a(4)=3$; thereafter $a(n) = a(n-1) + a(n-3)$.	$a(N) = a(N-1) + a(N-3)$
A173862	$a(n) = A158772(n-1)/21$.	$2^{\lfloor (N+1)/3 \rfloor}$
A000045	Fibonacci numbers: $F(n) = F(n-1) + F(n-2)$ with $F(0) = 0$ and $F(1) = 1$.	$F(N+2)$
A000045	Fibonacci numbers: $F(n) = F(n-1) + F(n-2)$ with $F(0) = 0$ and $F(1) = 1$.	$F(N)$
A001609	$a(1) = a(2) = 1, a(3) = 4$; thereafter $a(n) = a(n-1) + a(n-3)$.	$a(N) = a(N-1) + a(N-3)$
A169985	Round ϕ^n to the nearest integer.	$\text{round}(\phi^N)$
A005251	$a(0) = 0, a(1) = a(2) = a(3) = 1$; thereafter, $a(n) = a(n-1) + a(n-2) + a(n-4)$.	$a(N) = a(N-1) + a(N-2) + a(N-4)$
A005251	$a(0) = 0, a(1) = a(2) = a(3) = 1$; thereafter, $a(n) = a(n-1) + a(n-2) + a(n-4)$.	$a(N) = a(N-1) + a(N-2) + a(N-4)$
A000045	Fibonacci numbers: $F(n) = F(n-1) + F(n-2)$ with $F(0) = 0$ and $F(1) = 1$.	$F(N+2)$
A008619	Positive integers repeated.	$\lfloor (N+1)/2 \rfloor + 1$
A004525	One even followed by three odd.	$(N + 1 - \text{eps})/2 + 1, \text{eps} = \text{ceil}(N/2) \bmod 2$
A000045	Fibonacci numbers: $F(n) = F(n-1) + F(n-2)$ with $F(0) = 0$ and $F(1) = 1$.	$F(N)$
A000045	Fibonacci numbers: $F(n) = F(n-1) + F(n-2)$ with $F(0) = 0$ and $F(1) = 1$.	$F(N+2)$
A178715	$a(n)$ = solution to the "Select All, Copy, Paste" problem: Given the ability to type a single letter, or to type individual "Select All", "Copy" or "Paste" command keystrokes, what is the maximal number of letters of text that can be obtained with n keystrokes?	$a(N) = 4 a(N-5)$ for $N \not\equiv 16$
A214282	Largest Euler characteristic of a downset on an n -dimensional cube.	—
A001405	$a(n) = \text{binomial}(n, \lfloor n/2 \rfloor)$.	$\text{binomial}(N+1, \lfloor (N+1)/2 \rfloor)$
A006498	$a(n) = a(n-1) + a(n-3) + a(n-4), a(0) = a(1) = a(2) = 1, a(3) = 2$.	$F(\lfloor N/2 \rfloor + 2) * F(\text{ceil}(N/2) + 2)$
A195971	Number of $n \times 1$ 0..4 arrays with each element x equal to the number its horizontal and vertical neighbors equal to 2,0,1,3,4 for $x=0,1,2,3,4$.	$F(m+1)^2$ ($N=2m$ even), $F(m) F(m+3)$ ($N=2m+1$ odd)
A008619	Positive integers repeated.	$\lfloor N/2 \rfloor + 1$
A000034	Period 2: repeat $[1, 2]$; $a(n) = 1 + (n \bmod 2)$.	$1 + (N \bmod 2)$
A007877	Period 4 zigzag sequence: repeat $[0,1,2,1]$.	period 4: 1, 0, 1, 2 for $N = 1, 2, 3, 0 \pmod{4}$

4 Three that are more than bookkeeping

4.1 Dying rabbits, and rule 28

Rule 28 is IIVD. Its `obc0` recurrent-class count and its $D_{\max}$ ladder are *term for term identical to rule 70's* over the whole computed range, $N = 1 \dots 17$, at both boundary conditions:

$$2, 3, 4, 6, 8, 11, 15, 20, 27, 36, 48, 64, 85, 113, 150, 199, 264$$

$$D_{\max} = 1, 2, 2, 2, 4, 4, 4, 8, 8, 8, 16, 16, 16, 32, 32, 32, 64 = 2^{\lfloor (N+1)/3 \rfloor}.$$

The two rules are reflection partners, so the agreement is unsurprising; it is nonetheless *measured*, for the reason given above.

The class count is [A023434](#)($N + 2$), “dying rabbits”: $a(n) = a(n - 1) + a(n - 2) - a(n - 4)$ — Fibonacci’s rabbits with a finite lifetime, the $-a(n - 4)$ being the deaths. For W_{28} this is not merely the right recurrence; it reads as the right *story*. Rule 28 inherits rule 156’s 01 wall grammar, which on its own gives Fibonacci and φ (and does, for W_{156} : see the table). What rule 28 adds is a D channel, a reset to $|0\rangle$, which destroys a wall after finitely many steps. A wall with a finite lifetime is precisely what turns Fibonacci’s recursion into the dying-rabbit one, and $\varphi = 1.618$ into $\rho = 1.325$.

We flag this as an *interpretation to be checked against the wall grammar in Tier-2*, not a result. What is a result is the arithmetic: the exact recurrence, the identification, and the fact that the sequence which falsifies the φ prediction is a named classical object with a b-file.

A second observation belongs here. For W_{28} the weakly connected component count and the recurrent-class count coincide (11, 15, 20, 27, ... for both), so one identification covers both observables. That is not true across the whole ρ -group: W_{29} and W_{71} are the two rules whose WCC count differs from their recurrent-class count, which is why `sectors.rule_point` records W_{70} ’s base as an “integer recurrence” at 1.32472 but W_{29} ’s as a “parity fit” at 1.21470. The recurrent-class series has the exact Padovan law; the WCC series has only a fit.

4.2 Select All, Copy, Paste, and the $4^{N/5}$

W_{156}/W_{198} ’s largest sector runs

$$6, 9, 12, 16, 20, 27, 36, 48, 64, 81, 108, 144, 192, 256, 324, 432, 576$$

over $N = 6 \dots 22$, with asymptotic growth $4^{N/5} = 1.319508^N$. This is [A178715](#) exactly, with no shift.

The entry is Rowell’s “Select All, Copy, Paste” problem (*J. Integer Seq.* **18** (2015), 15.10.7): given a budget of n keystrokes, and the ability to type one letter, or to select-all/copy/paste and thereby multiply the current text by a factor m at a cost of $m + 1$ keystrokes, what is the longest text obtainable? The answer maximises a product of factors whose *costs* sum to the budget, where factor m costs $m + 1$. Optimising $m^{1/(m+1)}$ over integers gives $m = 4$ at $4^{1/5} = 1.319508$, beating $m = 3$ ($3^{1/4} = 1.316$) and $m = 5$ ($5^{1/6} = 1.308$) — and the entry’s own recurrence $a(n) = 4a(n - 5)$ for $n \geq 16$ is the statement that the optimum is eventually all fours.

That is where our exponent comes from. The largest Krylov sector of W_{156} is a product over the gaps between frozen walls, each gap of length ℓ contributing a factor, and the wall itself costing sites; the optimum therefore has the same shape as the keystroke problem, with the same integer optimisation and the same answer. R18 derived the Fibonacci recurrence for the *number* of wall sets from the grammar; A178715 supplies the matching account of the largest *preimage*.

4.3 W_{108} 's frozen states, factorised by the entry itself

R19 derived

$$n_{\text{frozen}}(W_{108}, N) = F_{\lfloor N/2 \rfloor + 2} F_{\lceil N/2 \rceil + 2},$$

a factorisation over the two sublattices. The series is [A006498](#)($N + 2$), defined by $a(n) = a(n - 1) + a(n - 3) + a(n - 4)$ — and the entry's own formula section states the Fibonacci-product identity. So the sublattice factorisation is not special to this model; it is a known property of that sequence, and our derivation and the encyclopedia's agree.

W_{201} 's frozen count is [A195971](#)($N - 1$), whose stated combinatorial meaning (a count of $n \times 1$ arrays over 0..4 with a neighbour condition) has no evident relation to anything here. We record the match and claim nothing from it.

5 Two identifications that are new results

W_{105} : central binomial. The largest sector is

$$D_{\max}(W_{105}, N) = \binom{N+1}{\lfloor (N+1)/2 \rfloor} = \text{A001405}(N+1),$$

verified over $N = 6 \dots 22$. This is the maximal level set of a balanced counting statistic, which is exactly what R21 then confirms: W_{105} has a single conserved range-2 charge, the staggered domain-wall number S , and its level sets *are* the Krylov sectors. The central binomial coefficient is the largest of those level sets, so the closed form is not a coincidence but the symmetry showing through.

W_{150} : an Euler characteristic. $D_{\max}(W_{150}, N)$ is [A214282](#)($N + 2$), “largest Euler characteristic of a downset on an n -dimensional cube”, a unique hit. W_{150} 's single conserved charge is the total domain-wall number D (R21), whose level sets are again exactly its sectors, so $D_{\max}$ is the largest such level set. The two sequences agree over the full 17 terms and diverge from W_{105} 's central binomials at $N = 9$ (210 against 252) and $N = 13$ (3003 against 3432) — the two rules are close but not the same, and the encyclopedia separates them.

R21 §6 now derives both. W_{150} 's sector sizes are $\binom{N+1}{D}$ over *even* D and W_{105} 's are $\binom{N+1}{w}$ over $w \equiv \lceil N/2 \rceil \pmod{2}$ — the same row of Pascal's triangle cut into its two parity classes. W_{105} 's class always contains the centre, hence the central binomial; W_{150} 's misses it exactly at $N \equiv 1 \pmod{4}$, which is where the divergences above sit and where they will keep sitting ($N = 17, 21, \dots$). The identification came first and the derivation second, which is the useful direction for an encyclopedia to work in.

And one gap the derivation closed. With the closed form in hand it was obvious what W_{105} 's *sector count* had to be — the size of a parity class, a period-4 staircase — and it is [A004525](#)($N + 2$), “one even followed by three odd”. That row had been missing from the table.

6 What is not there

Four series were searched for and returned nothing:

rules	bc	series
$W_{28}, W_{70}, W_{157}, W_{199}$	pbc	$n_{\text{rec}} = 1, 3, 4, 3, 6, 6, 8, 11, 13, 18, \dots$
W_{29}, W_{71}	pbc	$n_{\text{rec}} = 1, 2, 3, 2, 5, 5, 7, 10, 12, 17, \dots$
$W_{28}, W_{70}, W_{157}, W_{199}$	pbc	$D_{\max} = 1, 1, 2, 1, 2, 4, 2, 4, 8, 4, \dots$
W_{29}, W_{71}	pbc	$D_{\max} = 2, 1, 2, 1, 2, 4, 2, 4, 8, 4, \dots$

The branches are listed separately because the ring splits a group that `obc0` holds together: at `obc0` all six rules share one class-count sequence, whereas on the ring $\{28, 70, 157, 199\}$ and $\{29, 71\}$ differ at every $N \geq 2$. Their $D_{\max}$ ladders differ only at $N = 1$, where the “ring” is a single site.

Absence is recorded rather than omitted, because “we searched and there is nothing” is a usable statement. It is also not the same as “structureless”: the unmatched `pbcc` ladder satisfies $a(N) = 2a(N - 3)$ exactly from $N = 5$, which `test_oeis` pins.

7 Two junk hits, so that they are not rediscovered

[A173862](#). The ρ -group’s `obc0` $D_{\max}$ ladder $2^{\lfloor (N+1)/3 \rfloor}$ has exactly one OEIS home, and it is “ $a(n) = A158772(n - 1)/21$ ”. The terms are right and the match is exact, but nobody should cite it: the citable object is the closed form, and the encyclopedia simply has no primary entry for $2^{\lfloor n/3 \rfloor}$. We list it in the table for completeness and use the closed form everywhere else.

The trivial periodics. W_{150} ’s frozen count matches [A000034](#) (“period 2: repeat 1, 2”), W_{105} ’s matches [A007877](#) (“period 4 zigzag 0, 1, 2, 1”), and W_{150}/W_{156} ’s sector counts match [A008619](#) (“positive integers repeated”). These are correct and completely uninformative — any period-2 or period-4 sequence matches something. They are kept because the shift is still a check on the closed form: W_{105} ’s zero at $N \equiv 1 \pmod{4}$, which R19 had to special-case in `paper_figures._tier1a`, is visible in A007877’s phase.

8 Reproduce

```
python -m qca_fragmentation.scaling.oeis --refresh --tables
python -m pytest qca_fragmentation/tests/test_oeis.py -q
```

`--refresh` is the only step that touches the network; everything else reads the cached entries in `analytics/`. The identifications, the partner-agreement checks and the unmatched list are written to `oeis_identifications.json` in the same directory.